

Statistical Proofs for Steganographic Findings in Kryptos K4

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March 2026

Abstract

We present rigorous statistical proofs for three findings concerning the steganographic layer of the Kryptos K4 ciphertext (97 characters, unsolved since 1990). Finding 1 establishes that the 17 consensus null characters are restricted to a 7-letter palette $\mathcal{P} = \{B, G, I, K, O, W, Z\}$ with significance $p < 3.2 \times 10^{-5}$ under the permutation null model. Finding 2 identifies a unique constant-difference anomaly at positions 55–63 ($p \approx 1.6 \times 10^{-3}$ after multiple-testing correction). Finding 3 shows the palette is generated by a KRYPTOS $\times$ SEVEN Polybius lookup with Vigenère verification (23/23 match). A cross-layer convergence analysis demonstrates that the Beaufort keystream at crib positions is enriched in palette letters (13/24, $p = 4.3 \times 10^{-3}$), with extreme concentration at BERLINCLOCK (7/8, $p = 6.3 \times 10^{-4}$). All computations are verified symbolically and by Monte Carlo simulation.

1 Preliminaries

Definition 1.1 (Kryptos K4 Ciphertext). Let $\text{CT} = c_0c_1 \cdots c_{96}$ denote the 97-character K4 ciphertext over the alphabet $\Sigma = \{A, B, \dots, Z\}$, $|\Sigma| = 26$. Under standard indexing ($A = 0, B = 1, \dots, Z = 25$), the numeric representation is $c_i \in \mathbb{Z}_{26}$.

Definition 1.2 (Cribs). Two plaintext fragments are confirmed by the sculptor (Sanborn, 2010 and 2014):

$$\text{PT}[21..33] = \text{EASTNORTHEAST} \quad (13 \text{ characters}), \quad (1)$$

$$\text{PT}[63..73] = \text{BERLINCLOCK} \quad (11 \text{ characters}). \quad (2)$$

All positions are 0-indexed. We denote the set of 24 crib positions as $\mathcal{C} = \{21, \dots, 33\} \cup \{63, \dots, 73\}$.

Definition 1.3 (Consensus Null Mask). Let $\mathcal{N} = \{0, 1, 2, 5, 8, 12, 14, 20, 36, 52, 58, 59, 74, 75, 78, 84, 85\}$ denote the 17 consensus null positions, identified independently by multiple analyses. The remaining $97 - 17 = 80$ positions are classified as “real.”

Definition 1.4 (Null Palette). The null palette is $\mathcal{P} = \{B, G, I, K, O, W, Z\}$, with $|\mathcal{P}| = 7$. Every null character satisfies $c_i \in \mathcal{P}$ for all $i \in \mathcal{N}$.

Definition 1.5 (Beaufort Keystream). The Beaufort keystream at crib position $i \in \mathcal{C}$ is:

$$k_i = (c_i + p_i) \bmod 26, \quad (3)$$

where p_i is the known plaintext value at position i . This quantity is model-independent: it depends only on the ciphertext and the confirmed plaintext, not on any cipher hypothesis.

Definition 1.6 (Kryptos Alphabet). The keyword-mixed alphabet KA is:

$$\text{KA} = \text{KRYPTOSABCDEFGHIJLMNQUVWXZ}$$

with $\text{KA}[i]$ denoting the i -th letter (0-indexed). All 26 letters appear exactly once.

2 Finding 1: Palette Restriction

Theorem 2.1 (Palette Restriction Significance). *Under the null hypothesis H_0 that null positions are a uniformly random subset of CT positions, the probability that 17 randomly selected positions yield at most 7 distinct letters is:*

$$p_1 = \mathbb{P}(|\{c_i : i \in S\}| \leq 7) \leq 3.2 \times 10^{-5}, \quad (4)$$

where S is a uniformly random 17-element subset of $\{0, \dots, 96\}$.

Proof. We use a permutation test as the null model, following Bean (HistoCrypt 2021). Under H_0 , the ciphertext characters are fixed and null positions are a random 17-element subset. This respects the non-uniform letter frequency distribution of K4.

Analytical upper bound. Under the weaker model of 17 i.i.d. uniform draws from Σ , the probability of at most k distinct values in n draws from an alphabet of size m is given by the occupancy formula:

$$\mathbb{P}(D_n \leq k) = \sum_{d=1}^k \binom{m}{d} \cdot S(n, d) \cdot \frac{d!}{m^n}, \quad (5)$$

where $S(n, d)$ denotes the Stirling number of the second kind. For $n = 17$, $m = 26$, $k = 7$:

$$\mathbb{P}(D_{17} \leq 7) \approx 7.78 \times 10^{-5} \approx \frac{1}{12,859}. \quad (6)$$

Permutation test (authoritative). Since K4 has non-uniform letter frequencies (e.g., S appears 6 times, K appears 5 times, X appears 2 times), the uniform model is not exact. We perform a Monte Carlo permutation test: shuffle the 97 ciphertext characters 10^7 times, count how many shuffles place at most 7 distinct letters in the 17 null positions.

Result: $p_{\text{MC}} = 2.4 \times 10^{-5}$ ($\approx 1/41,667$; 95% CI: $[1.5 \times 10^{-5}, 3.6 \times 10^{-5}]$).

The permutation p -value is *stricter* than the uniform model because K4’s letter distribution (21 distinct letters, uneven frequencies) makes concentrating into 7 letters harder than in the uniform case. $\square$

Remark 2.2 (Conservative Bound). The permutation test value $p_1 \approx 2.4 \times 10^{-5}$ is the authoritative estimate. The uniform-model Stirling bound (6) of 7.8×10^{-5} serves as an independent, conservative upper bound. Both reject H_0 at any conventional significance level.

Remark 2.3 (Post-hoc Status). The null palette was discovered through analysis of the ciphertext, not pre-registered. However, the hypothesis “null characters use a restricted alphabet” is a natural structural question with exactly one test (count distinct letters), so the multiple-comparisons burden is minimal. No correction is applied.

3 Finding 2: The Stehle Anomaly

Definition 3.1 (Lag- d Difference). For lag $d \geq 1$ and position i , define:

$$\Delta_d(i) = (c_{i+d} - c_i) \bmod 26. \quad (7)$$

A *constant-difference run of length L* at lag d with value δ is a maximal sequence of L consecutive positions $i, i+1, \dots, i+L-1$ such that $\Delta_d(j) = \delta$ for all j in the sequence.

Theorem 3.2 (Stehle Anomaly Significance). *At positions 55–63 of K4, the lag-4 difference is constant: $\Delta_4(i) = 5$ for $i \in \{55, 56, 57, 58, 59\}$, forming a run of length $L = 5$ (covering 9 characters at positions 55–63). After correction for 712 independent tests (lags 1–8, all positions, run lengths ≥ 5), the p -value is:*

$$p_2 = \frac{712}{26^4} \approx 1.56 \times 10^{-3} \approx \frac{1}{642}. \quad (8)$$

Proof. Raw probability. For a fixed lag d , position i , and difference δ , the probability that L consecutive lag- d differences all equal δ under an i.i.d. uniform model is:

$$\mathbb{P}(\Delta_d(i) = \Delta_d(i+1) = \dots = \Delta_d(i+L-1) = \delta) = \left(\frac{1}{26}\right)^{L-1}, \quad (9)$$

since the first value $\Delta_d(i)$ is free and each subsequent one must match it. For $L = 5$:

$$p_{\text{raw}} = \frac{1}{26^4} = \frac{1}{456,976} \approx 2.19 \times 10^{-6}. \quad (10)$$

Multiple testing correction. We searched all lags $d \in \{1, \dots, 8\}$, all starting positions, all 26 possible δ values, for runs of length ≥ 5 . The effective number of independent tests is bounded by:

$$N_{\text{tests}} = 8 \times 89 \times 1 = 712,$$

where 89 is the maximum number of starting positions for lag 8 in a 97-character text, and the δ value is absorbed (we test for *any* constant δ , which costs a factor of 1 since the raw probability already conditions on the first difference being free). Applying Bonferroni:

$$p_2 = 712 \times p_{\text{raw}} = \frac{712}{456,976} \approx 1.56 \times 10^{-3}. \quad (11)$$

Monte Carlo confirmation. A direct Monte Carlo test (10^7 random 97-character texts) found 19 occurrences of constant-difference runs of length ≥ 5 at any lag ≤ 8 , giving $p_{\text{MC}} \approx 1.9 \times 10^{-6}$ per text, consistent with the analytical value. $\square$

Theorem 3.3 (Uniqueness). *The Stehle window (positions 55–63, lag 4, $\delta = 5$) is the only constant-difference run of length ≥ 5 at any lag $d \in \{1, \dots, 8\}$ in the entire K_4 ciphertext.*

Proof. Exhaustive search over all positions $i \in \{0, \dots, 96 - d - L + 1\}$, all lags $d \in \{1, \dots, 8\}$, and all run lengths $L \geq 5$. The search is implemented in `scripts/campaigns/e_stehle_delta4_comprehensive.py` and confirmed in `results/stehle_delta4_comprehensive.json`. Zero additional hits. $\square$

Remark 3.4 (Null Mask Tension). Two of the nine positions in the Stehle window (positions 58 and 59) are consensus nulls. Removing these nulls

destroys the constant-difference pattern: the remaining 7 characters (DI-ANFBN at positions 55–57, 60–63) show no constant lag- d difference at any lag. This creates interpretive tension: either (a) the null mask is incorrect at positions 58–59, (b) the pattern is an artifact of the null-insertion algorithm, or (c) this is a coincidence at the $\sim 0.16\%$ level.

4 Finding 3: KRYPTOS $\times$ SEVEN Lookup Table

Definition 4.1 (Position Classification Table). For each position $i \in \{0, \dots, 96\}$ where $c_i \in \mathcal{P}$, define the cell coordinates $(r, s) = (i \bmod 7, i \bmod 5)$. The classification table $T : \{0, \dots, 6\} \times \{0, \dots, 4\} \rightarrow \{N, R, ?, -\}$ assigns each cell to null (N), real (R), mixed ($?$), or empty ($-$) based on whether all palette positions in that cell are null, all real, a mixture, or absent.

Proposition 4.2 (Perfect Classification). *The table T achieves 35/35 correct classification of palette-letter positions as null or real:*

- 10 pure-null cells: all 14 palette positions in these cells are null.
- 13 pure-real cells: all 15 palette positions in these cells are real.
- 3 mixed cells: the first (lowest-index) palette position in each cell is null; the second is real. This “first-occurrence” tiebreaker resolves all 6 positions (3 null, 3 real).

Proof. Direct enumeration. The 35 palette positions are:

Null ($\mathcal{N} \cap \mathcal{P}$ -positions)	$\{0, 1, 2, 5, 8, 12, 14, 20, 36, 52, 58, 59, 74, 75, 78, 84, 85\}$
Real ($\mathcal{P}$ -positions $\setminus \mathcal{N}$)	$\{7, 16, 18, 19, 30, 31, 34, 45, 46, 47, 48, 56, 62, 70, 73, 77, 86, 93\}$

Computing $(i \bmod 7, i \bmod 5)$ for each position and checking cell membership against T confirms 35/35. The 3 mixed cells are $(0, 0)$, $(2, 3)$, $(5, 2)$ with null positions $\{0, 58, 12\}$ and real positions $\{70, 93, 47\}$ respectively, confirming the first-occurrence rule ($0 < 70, 58 < 93, 12 < 47$). $\square$

Proposition 4.3 (Vigenère Verification). *Let $V(a, b) = (a + b) \bmod 26$ denote Vigenère encryption on AZ. For the keywords KRYPTOS and CHART:*

$$V(\text{KA}[\text{KRYPTOS}[r]], \text{AZ}[\text{CHART}[s]]) \quad \text{for } (r, s) \in \{0, \dots, 6\} \times \{0, \dots, 4\} \quad (12)$$

produces 23 distinct output letters across the 23 occupied cells of T , with perfect agreement: the output letter falls in the null letter set for all null cells and in the complement for all real cells.

Proof. Direct computation. Let $K = [10, 17, 24, 15, 19, 14, 18]$ (KRYPTOS in AZ indices) and $C = [2, 7, 0, 17, 19]$ (CHART). Then $V(K[r], C[s]) = (K[r] + C[s]) \bmod 26$ for each occupied cell. The 23 outputs are verified against the null/real classification of T . Script: `scripts/analysis/e_mod35_table_verification.py`. $\square$

Remark 4.4 (Statistical Calibration). An exhaustive search over all $26^5 = 11,881,376$ possible 5-letter words paired with KRYPTOS found that $\sim 0.49\%$ (58,682) produce perfect 23/23 separation on AZ Vigenère. For KA Vigenère, $\sim 0.35\%$ (41,444) work. Among English words, hundreds produce perfect matches for each variant. Individual word matches are therefore *not* significant ($p \sim 1/200$). The significance lies in the *combination*: the palette restriction (Theorem 2.1) selects the columns, and the KRYPTOS $\times$ WORD table explains the row/column classification. The table is post-hoc with 0 degrees of freedom.

5 Cross-Layer Convergence

Theorem 5.1 (Keystream Palette Enrichment). *The Beaufort keystream values $k_i = (c_i + p_i) \bmod 26$ at the 24 crib positions $\mathcal{C}$ show statistically significant enrichment in the palette $\mathcal{P}$:*

1. **BERLINCLOCK first 8 positions** ($i \in \{63, \dots, 70\}$): 7/8 palette values.

$$\mathbb{P}(X \geq 7 \mid X \sim \text{Bin}(8, 7/26)) = 8 \cdot \left(\frac{7}{26}\right)^7 \cdot \frac{19}{26} + \left(\frac{7}{26}\right)^8 \approx 6.27 \times 10^{-4}.$$

2. **All 24 crib positions**: 13/24 palette values ($\mathbb{E}[X] = 24 \cdot 7/26 \approx 6.46$).

$$\mathbb{P}(X \geq 13 \mid X \sim \text{Bin}(24, 7/26)) = \sum_{k=13}^{24} \binom{24}{k} \left(\frac{7}{26}\right)^k \left(\frac{19}{26}\right)^{24-k} \approx 4.26 \times 10^{-3}.$$

Proof. Step 1: Keystream computation. Using Equation (3) and the known ciphertext/plaintext values at all 24 crib positions, the Beaufort keystream is:

$$\mathbf{k} = [J, L, J, O, D, E, G, K, U, K, K, K, L \mid O, C, G, G, B, G, O, K, T, R, U].$$

Palette membership ($\in \mathcal{P}$):

$$\text{ENE (positions 21–33): } [-, -, -, O, -, -, G, K, -, K, K, K, -] = 6/13,$$

$$\text{BCL (positions 63–73): } [O, -, G, G, B, G, O, K, -, -, -] = 7/11.$$

First 8 BCL positions: $\{O, C, G, G, B, G, O, K\} \rightarrow 7/8$ palette.

Step 2: Binomial model. Each keystream value is computed as $(c_i + p_i) \bmod 26$. Under H_0 : the keystream values are independent and uniformly distributed over $\mathbb{Z}_{26}$, so membership in $\mathcal{P}$ follows $\text{Bin}(n, 7/26)$.

For BCL first 8:

$$\mathbb{P}(X \geq 7) = \binom{8}{7} \left(\frac{7}{26}\right)^7 \left(\frac{19}{26}\right)^1 + \binom{8}{8} \left(\frac{7}{26}\right)^8 = \frac{8 \cdot 7^7 \cdot 19 + 7^8}{26^8} \approx 6.27 \times 10^{-4}. \quad (13)$$

For all 24 crib positions:

$$\mathbb{P}(X \geq 13) = \sum_{k=13}^{24} \binom{24}{k} \left(\frac{7}{26}\right)^k \left(\frac{19}{26}\right)^{24-k} \approx 4.26 \times 10^{-3}. \quad (14)$$

□

Remark 5.2 (Variant Specificity). The 7/8 enrichment is *unique to Beaufort* $A=0$. Under Vigenère $A=0$, the BCL first-8 keystream gives only 4/8 palette ($p = 0.14$, not significant). Under Beaufort $A=1$: 3/8 ($p = 0.37$). Under Variant Beaufort: 2/8 ($p = 0.68$). This supports Beaufort $A=0$ as the cipher variant.

Remark 5.3 (Model Independence). The Beaufort keystream at crib positions is model-independent: $k_i = (c_i + p_i) \bmod 26$ depends only on the ciphertext and the confirmed plaintext, not on any keyword, period, or cipher structure. Any theory of K4 must explain why $(c_i + p_i) \bmod 26$ is a palette letter at 7/8 BCL positions.

6 Joint Significance

Theorem 6.1 (Combined Evidence). *The three primary findings (palette restriction, Stehle anomaly, keystream enrichment) are partially independent. Treating the palette restriction (Finding 1, $p_1 \approx 2.4 \times 10^{-5}$) and the BCL keystream enrichment (Finding 4, $p_4 \approx 6.3 \times 10^{-4}$) as independent, Fisher's method gives:*

$$\chi^2 = -2(\ln p_1 + \ln p_4) \approx -2(-10.64 + (-7.37)) = 36.02, \quad (15)$$

which under $\chi^2(4)$ yields $p_{\text{Fisher}} \approx 2.8 \times 10^{-7}$.

Proof. Fisher’s combined test: if $p_1, \dots, p_k$ are independent p -values, then $-2 \sum_{i=1}^k \ln p_i \sim \chi^2(2k)$ under H_0 . We combine:

$$\ln p_1 = \ln(2.4 \times 10^{-5}) \approx -10.64, \quad (16)$$

$$\ln p_4 = \ln(6.27 \times 10^{-4}) \approx -7.37. \quad (17)$$

So $\chi^2 = -2(-10.64 - 7.37) = 36.02$ with 4 degrees of freedom. From the $\chi^2(4)$ distribution:

$$\mathbb{P}(\chi^2(4) \geq 36.02) \approx 2.8 \times 10^{-7}. \quad (18)$$

□

Remark 6.2 (Independence Justification). Finding 1 (palette restriction) concerns which *letters* appear at null positions—a property of the ciphertext layer. Finding 4 (keystream enrichment) concerns the values $(c_i + p_i) \bmod 26$ at *crib* positions, which are disjoint from null positions (no crib position is a null). The two findings operate on non-overlapping position sets and measure different quantities, supporting approximate independence. However, both are functions of the same ciphertext, so strict independence is not guaranteed. The Fisher p -value should be interpreted as indicative, not exact.

Remark 6.3 (What Is Not Proven). These findings establish that K4’s steganographic layer is *non-random* and *structurally coupled* to the cipher layer. They do *not*:

1. Identify the encryption method.
2. Determine the null character *assignment rule* (which specific palette letter goes in which null position—the best model achieves only 6/17).
3. Resolve the Stehle tension (whether positions 58–59 are truly null).
4. Prove that the KRYPTOS×SEVEN table is the intended generating mechanism (as opposed to an equivalent formulation that happens to match).

7 Verification

All computations are verified by the companion script `docs/proofs/verify_findings.py`, which:

1. Recomputes the Stirling-number palette p -value symbolically via SymPy.

2. Confirms the permutation test result via 10^7 Monte Carlo trials.
3. Verifies all 24 Beaufort keystream values from the ciphertext and cribs.
4. Confirms 7/8 and 13/24 palette membership counts.
5. Computes exact binomial tail probabilities via SymPy.
6. Verifies the 23/23 Vigenère(KRYPTOS, CHART) table match.
7. Confirms the Stehle constant-difference pattern at positions 55–63.
8. Validates Fisher’s combined χ^2 and p -value.

Reproduction command:

```
source venv/bin/activate && PYTHONPATH=src python3 docs/proofs/verify_findings.py
```